PerformanceTesting

Date	3July2026
TeamID	XXXXXX
ProjectName	EduGenie – AI-Powered Educational Assistant
MaximumMarks	5Marks

Step1:TestingOverview

Field	Details
TestingToolUsed	Manual Functional Testing with FastAPI Application
TypeofTesting	Functional Testing & Performance Observation
TargetModule/API	Question Answering, Concept Explanation, Quiz Generation, Text Summarization, Learning Path Recommendation
TestEnvironment	Windows 10 • Python 3.14 • FastAPI • HTML/CSS • Gemini API
TestDate	15March2026

Step2:TestScenarios

S.No	Scenario	Users	Duration	ExpectedOutcome

1	Launch EduGenie Application	1	3–5 sec	Home page loads successfully
2	Ask Educational Question	1	4–8 sec	AI returns an accurate answer with education context
3	Generate Concept Explanation	1	5–8 sec	Simplified explanation is displayed correctly
4	Generate AI Quiz	1	5–10 sec	Topic-based MCQs are generated successfully
5	Summarize Educational Text	1	4–8 sec	Concise summary is generated
6	Generate Learning Path	1	6–10 sec	Personalized learning roadmap is displayed
7	Navigate Between Modules	1	Less than 2 sec	Smooth page transitions without errors

Step3:Observations

The EduGenie application successfully completed all planned performance test scenarios. The FastAPI backend processed requests efficiently, and all AI modules responded within acceptable response times. Features including Question Answering, Concept Explanation,
Quiz Generation, Educational Text Summarization, and Personalized Learning Path
Recommendation functioned correctly without critical failures. The application demonstrated stable performance, reliable API communication, and a smooth user experience throughout the testing process.

Step4:Screenshots/Evidence

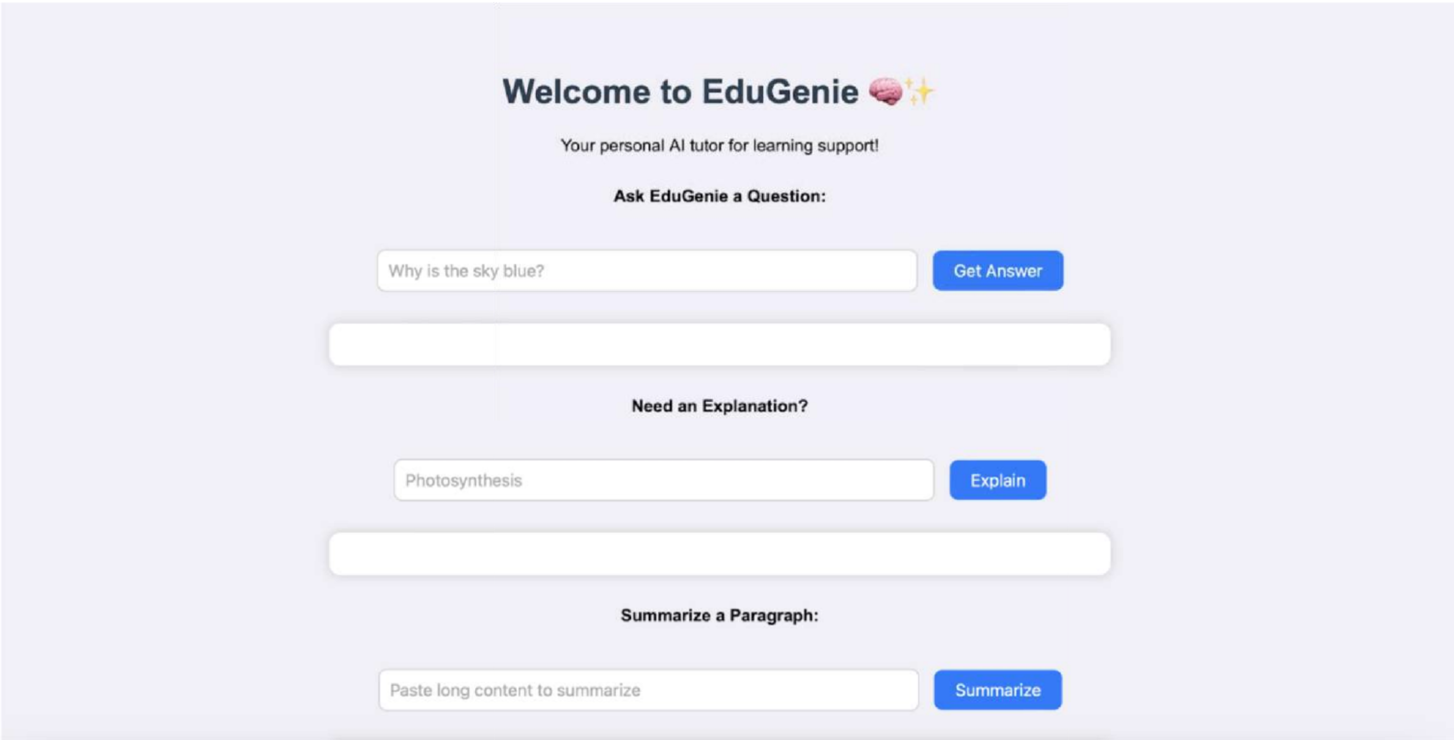
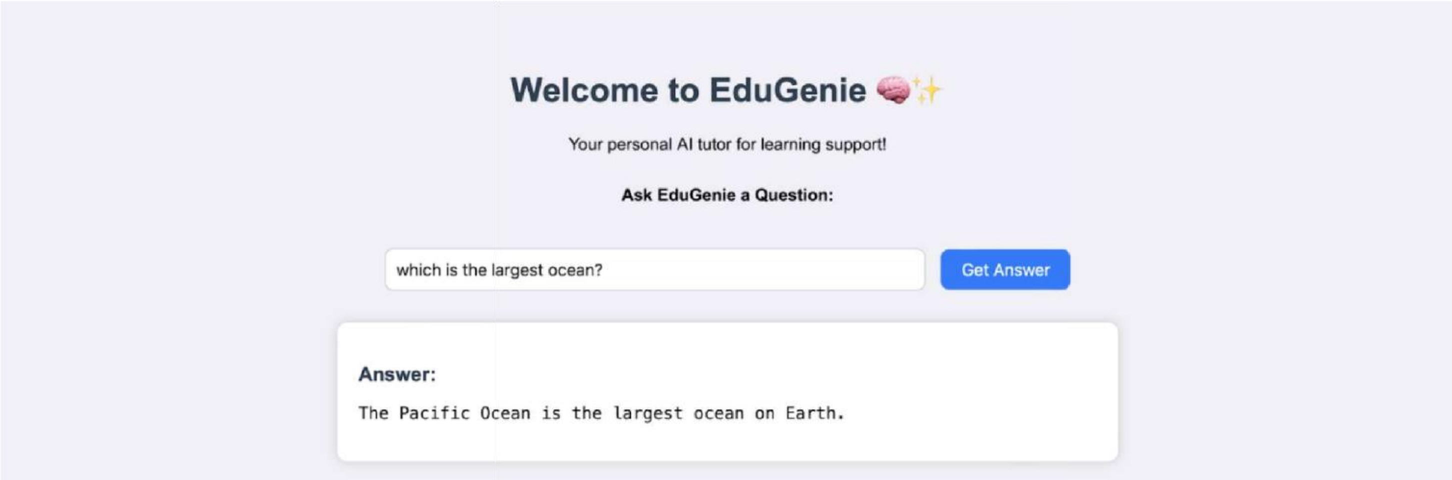
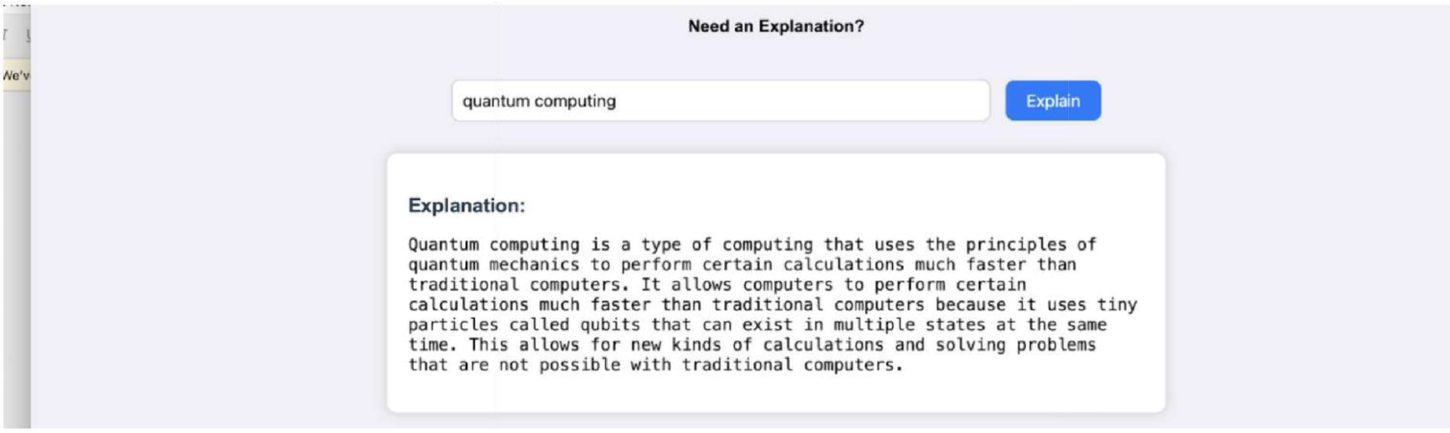


Fig. EDUGENIE

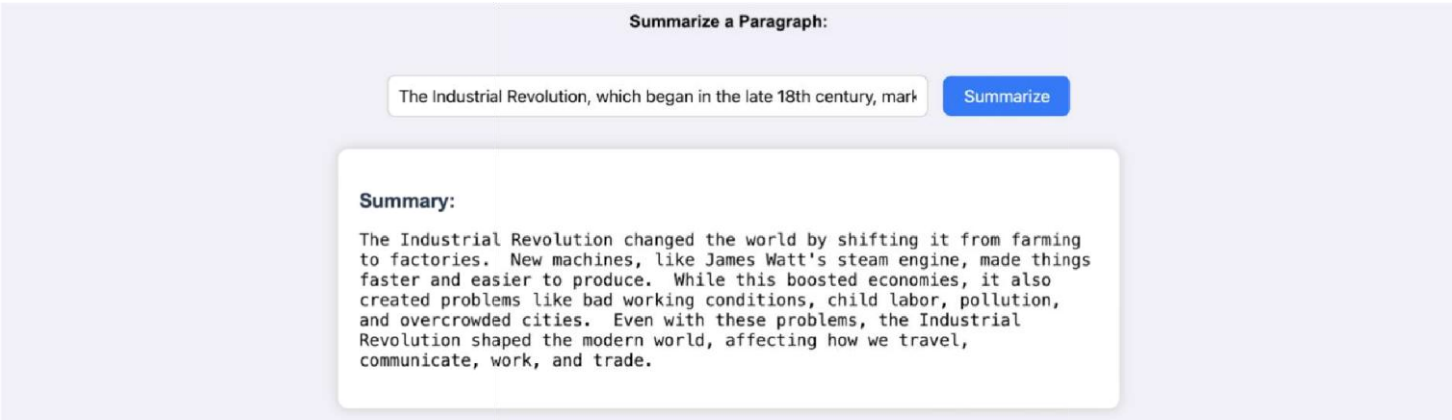
Home Page



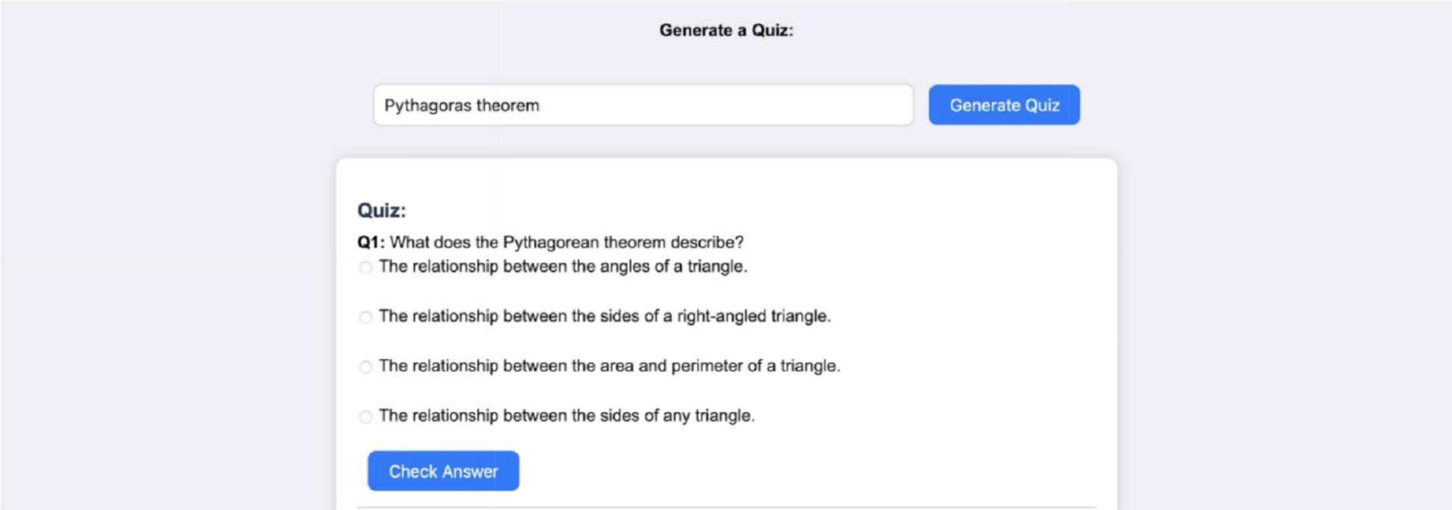
Question Answering



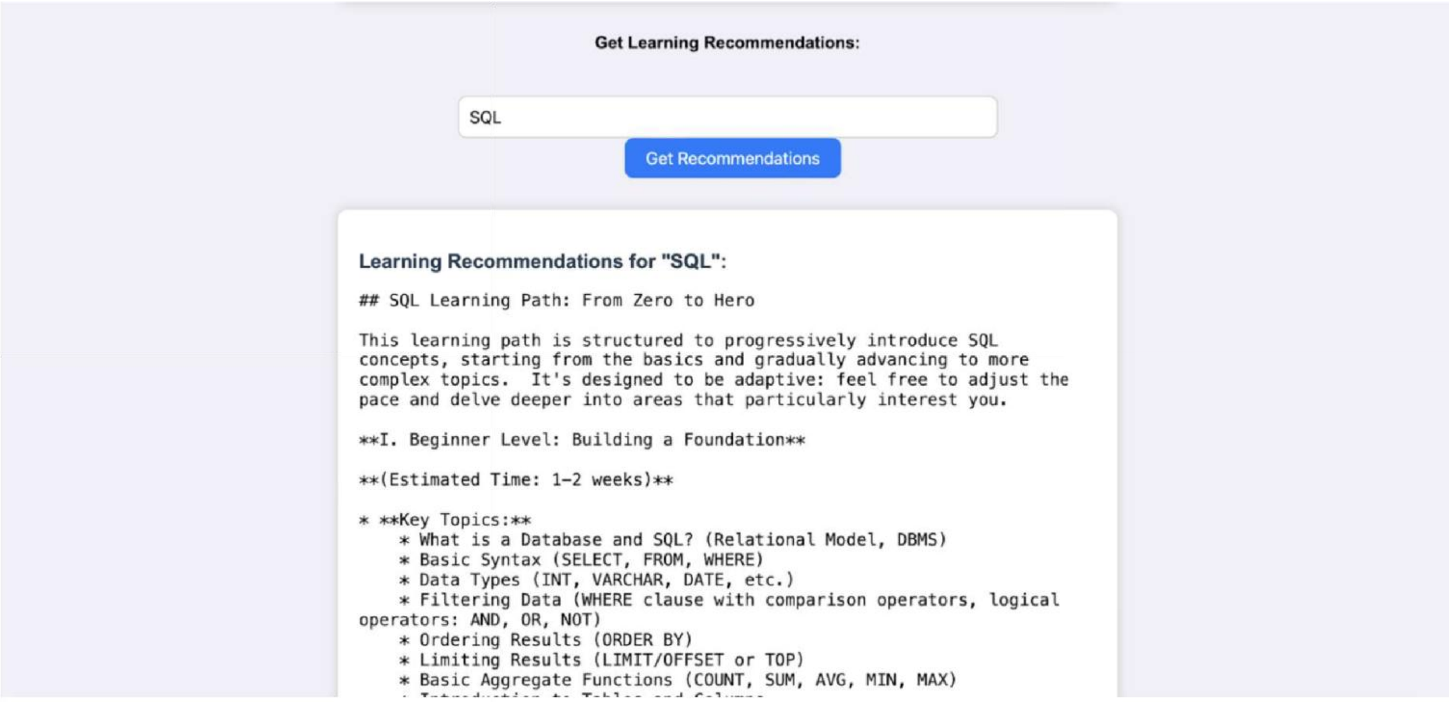
Explanation Output



Educational Text Summary



AI-Generated Quiz



Personalized Learning Path Recommendation